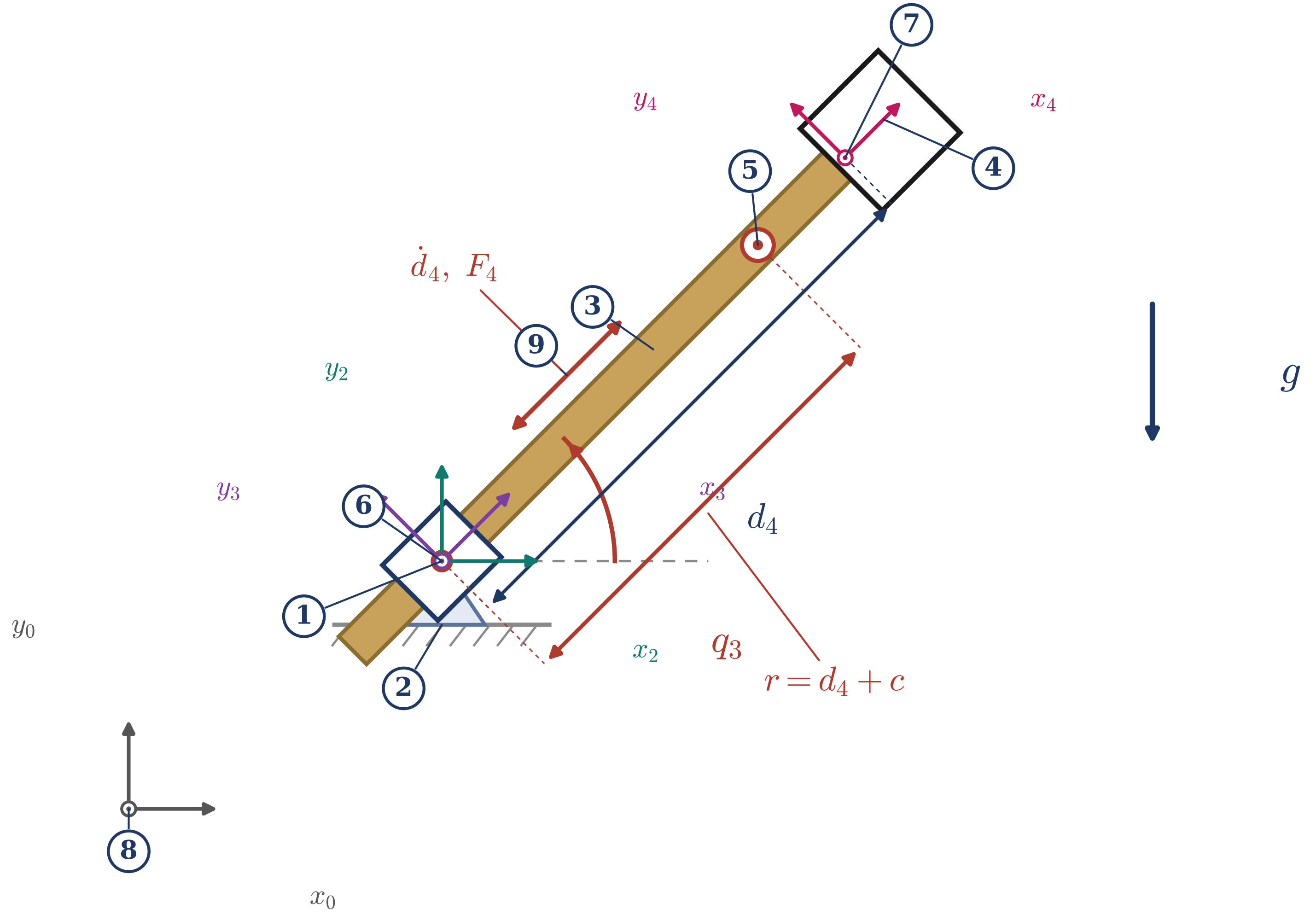
(b) Stage-1 abstraction: two degrees of freedom

q_3 saddle rotation (revolute) · d_4 crowd extension (prismatic) · boom welded to ground
swing and hoist-rope actuation enter at milestones M5–M6



Key

1 saddle pivot (revolute, q_3)

2 boom welded to ground (passive at Stage 1)

3 dipper handle (prismatic, d_4)

4 dipper

5 COM: $M_d = 54{,}300$ kg, $I_{zz} = 287{,}900$ kg m^2

6 $\{o_2\} \equiv \{o_3\}$ saddle and crowd frames coincide ($a_3 = 0$)

7 $\{o_4\}$ handle-end frame

8 $\{W\}$ world frame

9 crowd actuation (force along the handle)

$r = d_4 + c$ is the pivot-to-COM lever arm;
$c = -1.32$ m is a fixed bookkeeping constant.